

Some LED Lighting Players in India

- ▶ Abby Lighting & Switchgear
- ▶ Alien Energy
- ▶ Avni Energy Solutions
- ▶ Binay Opto Electronics
- ▶ C&S Electric
- ▶ Continental Device India
- ▶ GE
- ▶ Goldwyn
- ▶ Havells India
- ▶ InstaPower
- ▶ Kwaliti Photonics
- ▶ MIC Electronics
- ▶ Moser Baer India
- ▶ Osram India
- ▶ Philips Electronics India
- ▶ Regnant
- ▶ REI Electronics
- ▶ Reiz Electrocontrols
- ▶ Surya Roshni
- ▶ VuLED
- ▶ Wipro Lighting

and classroom sessions, participants learn the basics of LED technology while building an understanding of LED system integration issues involving the electrical, optical and thermal characteristics of LEDs. Participants also learn how to measure and evaluate LEDs and LED systems, and have the opportunity to compare LED technologies from a variety of manufacturers. The programme culminates with the participants designing, building and evaluating their own LED fixtures.

Different roles available

LED lighting is challenging to design as it requires complex driver and control electronics. As an LED lighting designer, one needs to achieve a practical balance between various factors including thermal management, efficiency, dimming, light and colour output, power factor correction (PFC), and safety.

Fortunately, IC manufacturers are responding to new LED device technologies with LED driver solutions that provide the key capabilities. Engineers can find suitable driver ICs optimised for LED lighting applications and tuned to work with specific classes of LEDs while addressing a



range of specialised requirements. Available ICs offer switched-mode solutions ranging from charge pumps to single-ended primary-inductor converters, thermal monitoring features, capabilities to help engineers meet PFC standards, and dimming control options available through a simple pulse-width-modulation control pin, among others.

If an LED luminaire is made in a well designed way, it can give 50,000 hours of life. On the other hand, LED drivers are available with a life expectancy of only 25,000 hours. Thus enhancing the life expectancy of LED drivers is one area you may be expected to focus on as an R&D professional.

Highly integrated ICs for lighting control and driver electronics bring power management to lighting applications. As a control and driver IC designer, you will be aiming at integration and finding new ways to reduce the form factor. Providing extensive support tooling is also important to allow for the fastest and simplest design. NXP Semiconductors, Infineon, Texas Instruments and Recom are some LED driver solution providers.

The main function of the mechanical design engineers is to work within the engineering team to design metal and plastic enclosures and additional mechanical components, production assistance with the design of jigs and fixtures, and also management of the jigs and fixture manufacture.

The responsibilities of the lighting application support manager include

supply design proposal—especially architecture lighting proposal, technical support to segment sales, product/application training to internal staff and professional dealers—especially for architecture knowledge and lighting, and coordination with internal department to organise and conduct lighting seminars/workshops. The candidate needs to be a university graduate with major in architecture, interior design, industry design or related field, and have some experience in architecture lighting design.

As a sales engineer, you will be responsible for handling projects, which include site visits, architect/consultant meets, planning and making technical presentation, proposal making, etc. You will need to explore possibilities to extend product portfolio, while doing extensive analysis of product range, technology used and price structure.

Future outlook

It is expected that by 2021, the LED technology will penetrate 57 per cent of the lighting market, driven by the government procurement and large-scale penetration by the public sector. Private sector, due to its cost economics, will be the early adopter. By 2031, more than 80 per cent of the lighting market will be captured.

So with LED becoming the lighting of choice, a career in this industry will only have brighter prospects. ●

The author is an executive editor at EYF